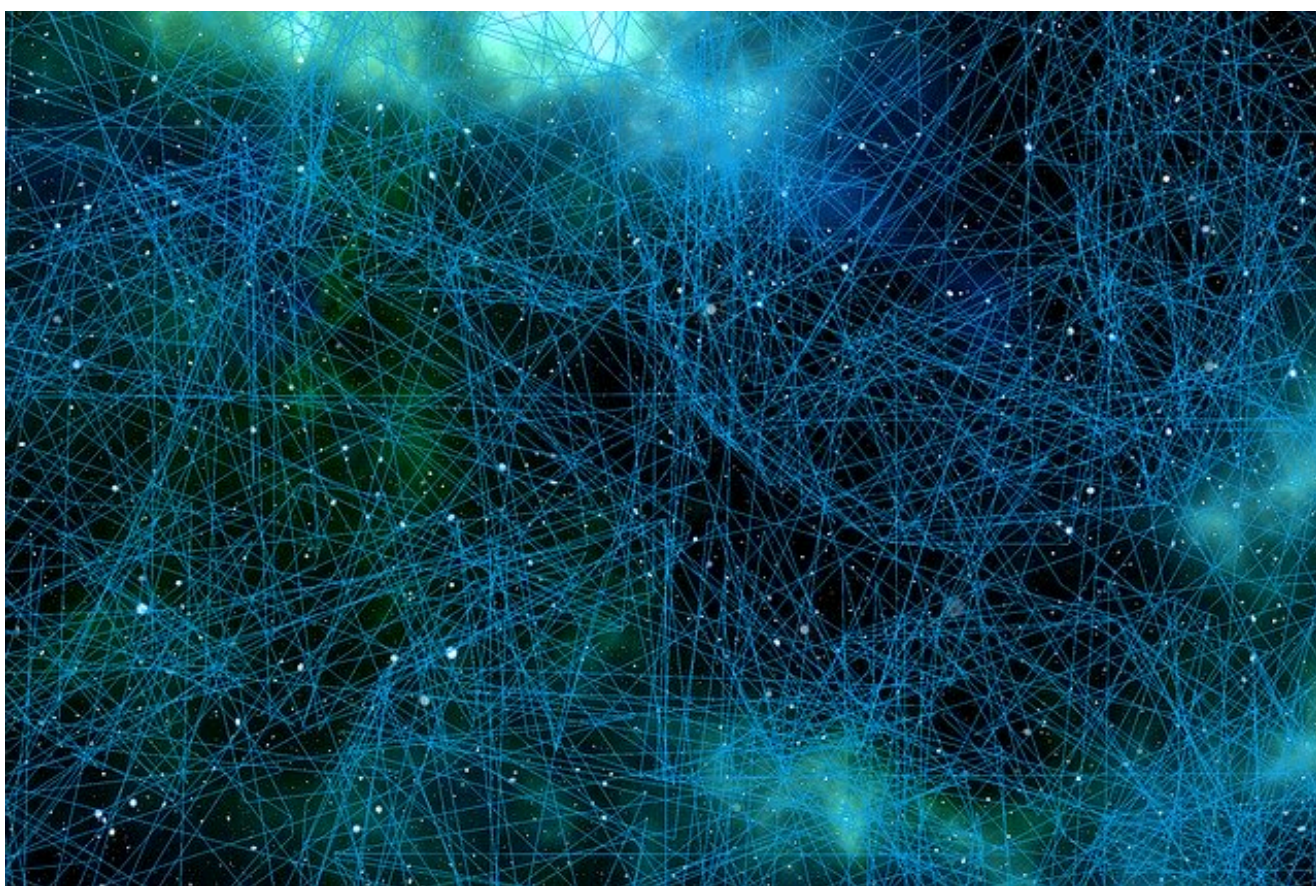


CLOUD NATIVE ECOSYSTEM / DEVOPS / МИКРОСЕРВИСЫ / СЕТЕВЫЕ ТЕХНОЛОГИИ

Stateless переосмысливает современные сетевые технологии

Stateless — это новый подход к сетевым технологиям для облачной эры.

16 марта 2020 года, 11:25 [Сьюзан Холл](#)



По мнению **Stateless**, компании из Боулдера, штат Колорадо, которая занимается **переосмыслением сетевых технологий**, с появлением новейших архитектур перед сетью открываются совершенно новые возможности.

Компания стремится упростить управление сетью, предоставляя платформу, которая может развиваться по мере изменения потребностей и масштабироваться по мере того, как ее клиенты — операторы колокейшн и поставщики сетевых услуг — будут решать задачи, связанные с 5G, Интернетом вещей и облачными технологиями.

Stateless Luxon, которое обеспечивает более глубокую программируемость коммутатора. В июне прошлого года Intel приобрела компанию Barefoot Networks, создателя независимого от протокола P4 языка программирования для сетевых устройств и семейства ASIC-коммутаторов Ethernet Tofino.

«Чтобы сохранить конкурентное преимущество, компаниям нужны быстрые и простые подключения к облаку, поддержка многочисленных межсоединений и возможность быстро развертывать разнообразные новые приложения. Сегодня для этого часто требуются сложные инженерные решения и новое оборудование, а на развертывание одного облачного приложения у компании могут уйти месяцы», — сказал соучредитель и генеральный директор **Мурад Каблан**. Stateless призван изменить ситуацию.

Клиенты компании стремятся обеспечить мультиарендность и многоуровневую мультиарендность — сотни тысяч пользователей в одном кластере. DevOps-команды могут захотеть создать сеть и подключение, а через пару часов удалить их, вместо того чтобы ждать шесть недель, пока ИТ-отдел их настроит. Клиентам нужна гибкая эластичная инфраструктура — облачная модель, позволяющая использовать ресурсы по мере необходимости и работать в тандеме со старыми технологиями, не прибегая к полной замене.

«В эпоху облачных технологий предприятия привыкли управлять инфраструктурой с помощью программного обеспечения, но сети относительно медленно перенимают эту тенденцию, — сказал старший аналитик 451 Research **Крейг Мацумото**. — Решение компании Stateless Inc. добавить технологию программируемых коммутаторов P4 в свою платформу Luxon обеспечивает высокоуровневое управление сетью и позволяет пользователям проще создавать программно-конфигурируемые сети».

Flexibility and Visibility

Kablan and **Eric Keller** launched Stateless in 2016 based on their research in the **Electrical, Computer and Energy Engineering Department** at the University of Colorado.

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Using its software-defined interconnect (SD-IX), Stateless enables colocation providers and data center operators to dynamically deliver routing, security and automation services to network interconnection points through software. It provides them the ability to control and connect every endpoint through APIs, including portfolio data centers, tenant sites and hyperscale clouds.

Stateless runs on commodity x86 servers along with the P4 switches and Luxon software.

It's been tricky to attain the right balance of hardware and software with networking. Traditional networking equipment relied on hardware for speed, though it was inflexible, according to **Simon Wheeler**, director of product management at Stateless. Meanwhile, **software-defined networking** provides flexibility, but is slower. Network functions virtualization (NFV) was software built like a hardware appliance, with all its associated limitations. And while it can run on commodity x86 servers, it requires tight control over specific types of hardware, he said.

"We went down the cloud native application approach using a microservices environment. In order to do this, we have to employ a scale-out cluster or a horizontally scalable cluster," Wheeler said.

This pools resources of compute and storage and RAM. That pooling of resources and the elastic usage of it helps mitigate the slowness of software and increase the speed, he said.

Yet there was still a hardware dependency involving SDN switches using **OpenFlow**, a multivendor standard defined by the **Open Networking Foundation** (ONF).

"What we realized was that we would have to program OpenFlow all the way, really down to the ASIC, and we weren't able to do that. And the flexibility that we needed, wasn't there," he said. A number of its colocation provider customers are interested

The Tofino switch makes the application-specific integrated circuit (ASIC) itself programmable. The forwarding logic resides in the P4 program that is loaded onto the chip. No protocol is baked into the chip. When a new protocol is required, the operator can simply add new logic to the P4 program.

“It gives us a far greater capability and flexibility through software to add new features, remove unused protocols, but also critically for the **networking industry**, to give greater visibility of traffic as it goes through the switch itself,” Wheeler said.

With Luxon, it’s able to tag packets, for instance, as they pass through this switch, providing enhanced visibility and capability to enforce quality-of-service limits, “which is something that has been unheard of, especially for colocation providers,” he said.

“With P4, Stateless can perform very fine-grained control of packet flows, including load balancing, QoS, both across network functions, and across network tenants,” he wrote in a **blog post**.

“P4 and Tofino have given Stateless the tools we needed to reinvent network functions from the very bottom up, including both network function architecture and the control and data plane adaptations to take full advantage of the capabilities of modern software design.”

Decoupling from State

Stateless holds a patent for more efficient package routing that involves four key elements: a high-speed network interface controller for fast read/write of packets, the distributed storage technique **RAM Cloud** for rapid state lookup, batched parallel processing for state lookups; and distributed state storage in multiple locations in multiple servers.

Decoupling from state is central to the Stateless approach. Rather than each network function having to keep track of all the traffic, it uses a distributed data store that provides that information to stateless network functions through a microservices architecture. The functions can then be decomposed into individual components that can be operated through orchestration. That enables clients to provide a platform where tenants can define their custom networks while the work launching and managing the interconnections is automated behind the scenes.

At GA last September, the company announced interconnect service chains that enables customers to mix and match any of the service functions, including security services like packet filtering or encryption or different versions of routing.

“New interconnect requirements driven by emerging network applications and functions are causing networks to undergo tectonic shifts in how they are designed, built and operated,” said **Arkadiy Shapiro**, product line manager at Barefoot Networks. “Stateless’s move to combine the power of its software-defined interconnect platform with the agility of Barefoot Tofino’s P4 programmability **elevates networking capabilities** and offers users a modern, fully-programmable network functions platform.”

Image by Gerd Altmann from Pixabay.

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